

CSA07-CN_C03-AT2-Case Study

Name: R Lakshana
Reg No: 1972260027

Q1. RIP — Chennai HQ Campus

Brief Problem Statement:

In March 2024, NimbusEdge Technologies added a new subnet for the Delivery department at the Chennai HQ campus. This change caused the network to exceed RIP v2's maximum reachable distance of 15 hops. The new Delivery subnet therefore reached a RIP metric of 16, which RIP treats as unreachable. The route was silently removed/excluded from routing, causing connectivity problems. No automatic alert was generated, so the issue was discovered only after client support tickets were raised.

A) Understanding of the Case

During March 2024, NimbusEdge added a new subnet for the Delivery department at the Chennai HQ campus.

The important issues were:

1. Network change

The addition of the new Delivery subnet caused the routing path to exceed the maximum hop-count supported by RIP v2.

2. Technical trigger

RIP uses hop count as its routing metric. RIP can consider routes reachable only up to 15 hops. The new Delivery subnet became a 16-hop route.

Therefore:

15 hops → reachable ;; 16 hops → unreachable

3. Effect on the Delivery subnet

Once RIP considered the route to have a metric of 16, the route was treated as unreachable and was not maintained as a valid route in the routing table. As a result, devices relying on that route could not reach the Delivery subnet.

4. Why it was not detected immediately

The incident report states that no alert was raised because RIP does not treat an unreachable network beyond its hop limit as a configuration error requiring an automatic alert. The routing protocol therefore followed its normal rules and silently discarded the unusable route.

5. How the problem was discovered

The NOC engineers did not identify the problem immediately. It became visible when client support tickets started appearing because users were experiencing connectivity problems.

Thus, the incident involved both: Network change → 16-hop route → RIP considers route unreachable → route dropped → no alert → client tickets → engineers discover outage

B) Application of RIP v2 Concepts —

RIP v2 is a distance-vector routing protocol that uses hop count to determine route reachability.

The failure can be explained step by step.

Step 1: Delivery subnet is introduced : NimbusEdge adds the new Delivery department subnet to the Chennai HQ network.

Step 2: RIP advertises the subnet : RIP routers exchange routing information with their neighboring routers. The hop count represents the distance to the destination.

Step 3: Hop count increases : As the route passes through routers, the RIP metric increases by one for each hop.

The important limit is: Maximum reachable RIP metric = 15

Step 4: Route reaches 16 hops : Because of the network change, the Delivery subnet becomes a 16-hop destination.

At this point, RIP does not treat it as a normal reachable route.

Step 5: RIP assigns infinity

RIP defines:

Metric 16 = Infinity / Unreachable

Therefore, the 16-hop Delivery subnet is considered unreachable.

Step 6: Route is excluded : Since the metric is 16, the router does not install/retain the Delivery subnet as a usable route in its routing table.

Step 7: Traffic cannot use the route : Without a valid routing-table entry, traffic destined for the Delivery subnet cannot be forwarded through that RIP route.

RIP metric	Meaning
1-14	Reachable route
15	Maximum reachable hop count
16	Unreachable / Infinity

Therefore, the key reason for the failure is that the Delivery subnet crossed RIP's 15-hop limit, reaching the unreachable value of 16.

c) Analysis

The absence of an automatic alert was not simply a one-time human error. It exposed a structural limitation of using RIP as a routing protocol. RIP's primary function is to exchange routing information and calculate routes using hop count. It does not inherently provide complete end-to-end service monitoring. In this incident, RIP correctly followed its protocol behavior:

Route metric reaches 16 → route considered unreachable → route removed/excluded

However, this protocol behavior did not automatically mean that the NOC monitoring system would generate an alarm.

Monitoring gap : The RCA monitoring gap is important because the infrastructure could technically be operating according to RIP's rules while a business-critical subnet was unreachable.

The monitoring system should have detected conditions such as:

Loss of the Delivery subnet route, Unreachable critical subnet , Abnormal routing-table changes , Failure of connectivity to important department networks , Instead, the problem remained unnoticed until clients reported it , Structural limitation.

This demonstrates that routing correctness and service monitoring are different things.

RIP can determine that a route is unreachable, but additional monitoring is required to tell network administrators that this routing event is affecting an important service or subnet.

Therefore, the incident reveals a systemic weakness:

RIP route failure + insufficient monitoring = silent network outage

The human error was the network change, but the prolonged detection time was caused by the monitoring gap.

D) Solution Design

Recommended Solution: Migrate Chennai HQ to OSPF

For the Chennai HQ, a justified long-term solution would be to migrate from RIP v2 to OSPF if the network is expected to grow or contain more routed segments. OSPF does not have RIP's 15-hop limitation. It is therefore more suitable when the number of routed segments increases.

Why OSPF is suitable

No 15-hop limitation: The specific problem that caused this incident would not occur because of RIP's 15-hop maximum.

Better scalability: OSPF is designed for larger and more complex IP networks.

Better route calculation: OSPF uses link-state information and cost rather than simply counting router hops.

Faster convergence: OSPF generally provides faster convergence than traditional distance-vector routing protocols such as RIP.

Future growth: If Chennai HQ adds more departments and subnets, OSPF provides greater room for expansion.

Feasibility for Chennai HQ:

The migration can be performed in a controlled manner:

Existing RIP network → configure OSPF → test routes → migrate during maintenance window → verify all department subnets → remove RIP

Critical Delivery and other department subnets should be tested before and after the migration. Monitoring should also be improved so that loss of an important subnet generates an immediate NOC alert.

Why RIP Was Originally a Reasonable Choice:

It is important not to conclude that RIP was a completely inappropriate choice.:

The incident describes the Chennai HQ segment as a small and stable campus network. RIP can be reasonable in such an environment because: It is simple to configure, It is easy to maintain, It has relatively low configuration complexity, A small, stable network may not require the complexity of OSPF, Network administrators can easily understand its hop-count metric. The problem occurred when the network was changed and the topology exceeded RIP's fundamental limitation. Therefore, RIP was reasonable for the original small and stable environment, but the addition of the new subnet created a condition where its 15-hop limitation became unacceptable.

E) Clarity

Technical Reasoning:

The Delivery subnet was added during March 2024. The resulting routing distance exceeded RIP's maximum reachable value of 15 hops.

The route therefore became:

Delivery subnet → 16 hops → RIP metric 16 → Infinity/Unreachable → route excluded

Because no suitable monitoring alert existed, the NOC did not immediately know that the subnet had become unreachable. Client support tickets eventually exposed the problem.

Recommendation:

NimbusEdge should: Review the Chennai HQ routing topology, Avoid designs that approach RIP's 15-hop limit if RIP is retained, Add monitoring for critical subnet reachability and route loss, Prefer OSPF if Chennai HQ is expected to grow or become more complex , Perform proper testing after every routing/topology change.

Conclusion:

The March 2024 Chennai HQ incident was caused by the interaction of a network change, RIP's 15-hop limitation, and a monitoring gap. The Delivery subnet reached a RIP metric of 16, which represents infinity/unreachable, so the route was excluded from the routing table.

The incident also demonstrates an important networking lesson: a routing protocol can behave correctly according to its rules while the network service is still unavailable. Therefore, routing protocols must be supported by proper monitoring and change-management procedures.

For the original small and stable Chennai HQ environment, RIP was a reasonable choice because of its simplicity. However, after the network expanded beyond its practical limits, OSPF would be a more suitable long-term solution.

Q2. IGRP — Bengaluru–Pune–Hyderabad Regional Link

a) Understanding of Case

The Bengaluru–Pune regional link experienced an **ISP-side fault**, which degraded the link capacity to around **2 Mbps**. A healthier alternate route was available through **Pune–Hyderabad–Bengaluru**. However, the network continued using the degraded Bengaluru–Pune path for some time.

Because of the reduced bandwidth, packets experienced **higher queuing and transmission delay**, causing increased latency. This affected business applications and resulted in noticeable performance problems.

b) Application of IGRP Composite Metric

IGRP uses a composite metric based on factors such as bandwidth and delay. Using the simplified formula:

$$\text{Metric} = 10^7 / \text{Bandwidth(kbps)} + \text{Delay}$$

For the degraded Bengaluru–Pune link:

$$2 \text{ Mbps} = 2000 \text{ kbps}$$

$$\text{Metric} = 10^7 / 2000 + \text{Delay}$$

$$\text{Metric} = 5000 + \text{Delay}$$

Thus, the reduced bandwidth makes the IGRP metric much higher.

If the alternate route has a **lower composite metric**, IGRP should select it instead of the degraded Bengaluru–Pune route.

Therefore, IGRP should have recalculated the metrics after the link degradation and shifted traffic toward the healthier alternate path.

c) Analysis

The main issue was **slow convergence rather than an incorrect metric calculation**.

IGRP's metric calculation could identify that the 2 Mbps link had become less attractive. However, IGRP required time to detect the degradation, exchange routing updates, recalculate the route and install the new path.

During this convergence period, traffic continued to use the degraded link, resulting in high latency.

Therefore:

Incident = Convergence-speed issue

d) Solution Design: NimbusEdge can improve the situation by **tuning IGRP timers** so that routing changes are detected and propagated faster. However, for a regional network, a better long-term solution is to **migrate to a faster-converging protocol such as OSPF or EIGRP**.

This provides:

- Faster route convergence.
- Better scalability.
- Faster recovery from link failures.
- More reliable routing for regional traffic.

Therefore, protocol migration is recommended for NimbusEdge's regional network.

e) IGRP Advantage Over RIP

RIP mainly selects routes using **hop count**, while IGRP uses a **composite metric involving bandwidth, delay, reliability and load**.

Therefore, IGRP was originally expected to make better routing decisions than RIP because it could prefer a **higher-quality path**, rather than simply choosing the route with fewer hops.

However, its advantage failed during this incident because the routing information did not reconverge quickly enough after the real-time link degradation.

Conclusion

The Bengaluru–Pune link degraded to **2 Mbps**, increasing its metric:

Metric = 5000 + Delay

IGRP should have recognized the degraded route and preferred the healthier Pune–Hyderabad–Bengaluru path. The main problem was **slow convergence**, which caused continued traffic on the degraded link and increased latency. Migrating to **OSPF/EIGRP** would provide faster and more reliable recovery.

Q3. OSPF — Core Backbone Migration

a) Understanding of Case

The Hyderabad router's OSPF adjacency failed because of an **incorrect Area ID entered during a configuration change**. The router was placed in a different OSPF area from its neighboring router. Because the adjacency was not formed, routing information was not exchanged correctly. This created a **routing black hole**, where traffic was sent toward the affected network but could not reach its destination. Since the physical link remained up, standard monitoring did not immediately detect the problem.

b) Application of Concepts

For OSPF neighbors to form an adjacency, important parameters must match, including:

- **Area ID**
- Hello and Dead intervals
- Authentication settings
- Compatible subnet/network parameters

In this case, the **Area ID was mismatched**.

Before:

Hyderabad Router → Correct Area ID

After incorrect change:

Hyderabad Router → Wrong Area ID

Because the neighboring routers belonged to different OSPF areas, they rejected the Hello relationship and **failed to establish adjacency**.

c) Analysis

This fault was harder to detect than a physical link failure because the interface remained **physically up**, even though OSPF communication was broken.

The RCA also showed a **change-management gap**. A single transcription/configuration error was allowed to go live without proper validation or peer review. There was no effective post-change adjacency check, allowing the problem to remain undetected.

d) Solution Design

The technical solution is to **correct the Area ID** and verify that both routers have matching OSPF parameters.

A proper verification procedure should include:

- Check Area ID before applying changes.
- Use peer review for routing changes.
- Verify OSPF neighbor adjacency after configuration.
- Maintain an adjacency verification checklist.

NimbusEdge is migrating to **multi-area OSPF** because it provides:

- Faster convergence.
- Better scalability through hierarchical areas.
- Accurate link-state routing.
- Better control than RIP/IGRP.

e) Conclusion

The main problem was an **incorrect Area ID configuration** on the Hyderabad router. This prevented OSPF adjacency, causing a routing black hole while the physical link appeared normal.

Correct configuration, peer review, and post-change adjacency verification can prevent recurrence. Multi-area OSPF provides NimbusEdge with **faster, more scalable and accurate routing** than the older RIP/IGRP setup.

Q4. BGP — ISP Peering and Failover

a) Understanding of Case

NimbusEdge experienced a failure on **ISP-A**. The network was expected to automatically shift traffic to **ISP-B** through BGP failover.

However, the backup route was not selected because of an incorrect **BGP routing policy**. As a result, traffic was disrupted until network engineers manually changed the configuration and restored connectivity. The incident caused service interruption and ultimately resulted in a **formal SLA breach notice** from clients.

b) Application of BGP Concepts

BGP uses attributes such as **AS-PATH**, **LOCAL_PREF** and **MED** to select the preferred route.

A likely problem was an incorrect **LOCAL_PREF policy** that continued to prefer routes learned through ISP-A, even when ISP-A was unavailable or degraded.

Before:

ISP-A → Higher LOCAL_PREF → Preferred

ISP-B → Lower LOCAL_PREF → Backup

Because of this policy, automatic failover did not work as expected.

Correct policy:

ISP-A → Primary

ISP-B → Lower LOCAL_PREF → Backup

When ISP-A fails, its route should disappear and BGP should automatically select ISP-B.

c) Analysis

An untested BGP failover policy is risky because BGP controls **external/ISP route selection** and incorrect policies can affect large amounts of traffic.

Unlike an interior routing problem, a BGP policy error can cause traffic to be **dropped, misrouted or sent through an unsuitable ISP**.

The business impact was:

ISP-A failure → BGP failover failed → connectivity disruption → manual intervention → service delay → client impact → SLA breach.

d) Solution Design

NimbusEdge should configure BGP so that ISP-A has a higher **LOCAL_PREF**, while ISP-B has a lower value as the backup.

After ISP-A becomes unavailable:

ISP-A route withdrawn → ISP-B route selected → Traffic automatically switches to ISP-B.

The configuration should be tested using controlled failover tests. BGP policies should also be reviewed regularly and tested at scheduled intervals.

BGP is the correct choice for dual-ISP redundancy because it is designed for **inter-AS routing and ISP connectivity**. RIP, OSPF or IGRP are mainly interior routing protocols and are not suitable for managing external ISP path policies.

e) Conclusion

The main failure was an **incorrect or untested BGP policy** that prevented automatic failover from ISP-A to ISP-B.

A correct **LOCAL_PREF-based primary/backup policy**, combined with regular failover testing and peer review, would provide reliable ISP redundancy and help prevent future SLA breaches.